

It is a cold, windless morning and three hot-air balloons hover above a park. Each balloon is stationary and in direct line of sight, with no obstacles near them, as shown in Figure 3. Balloon A is equipped with a 100 W siren, which emits a 2000 Hz tone uniformly in all directions. On board balloons B and C are students with sound measuring equipment.

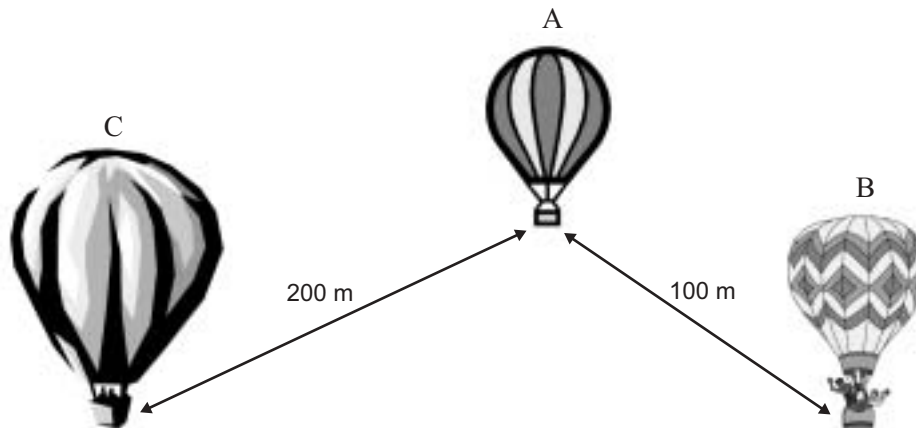


Figure 3

Question 9

Which of the following is the best estimate of the sound **intensity** of the siren as measured at balloon B?

- A. 0.5 W m^{-2}
- B. $2.5 \times 10^{-2} \text{ W m}^{-2}$
- C. $8.0 \times 10^{-4} \text{ W m}^{-2}$
- D. $2.5 \times 10^{-5} \text{ W m}^{-2}$

2 marks

Question 10

By how many decibels will the sound intensity level at balloon C be lower than at balloon B?

 dB

2 marks

Question 11

Balloons B and C move so that they are at equal distances from balloon A.

The sound intensity at balloon C is now measured as $1.0 \times 10^{-2} \text{ W m}^{-2}$.

What is the sound intensity level (dB) at balloon B?

 dB

2 marks

PHYSICS

Written examination 2

DATA SHEET

Directions to students

Detach this data sheet before commencing the examination.
This data sheet is provided for your reference.

1	photoelectric effect	$E_{k\max} = hf - W$
2	photon energy	$E = hf$
3	photon momentum	$p = \frac{h}{\lambda}$
4	de Broglie wavelength	$\lambda = \frac{h}{p}$
5	resistors in series	$R_T = R_1 + R_2$
6	resistors in parallel	$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2}$
7	magnetic force	$F = I l B$
8	electromagnetic induction	emf : $\varepsilon = -N \frac{\Delta \Phi}{\Delta t}$ flux: $\Phi = BA$
9	transformer action	$\frac{V_1}{V_2} = \frac{N_1}{N_2}$
10	AC voltage and current	$V_{\text{RMS}} = \frac{1}{\sqrt{2}} V_{\text{peak}} \quad I_{\text{RMS}} = \frac{1}{\sqrt{2}} I_{\text{peak}}$
11	voltage; power	$V = RI \quad P = VI$
12	transmission losses	$V_{\text{drop}} = I_{\text{line}} R_{\text{line}} \quad P_{\text{loss}} = I_{\text{line}}^2 R_{\text{line}}$
13	mass of the electron	$m_e = 9.1 \times 10^{-31} \text{ kg}$
14	charge on the electron	$e = -1.6 \times 10^{-19} \text{ C}$
15	Planck's constant	$h = 6.63 \times 10^{-34} \text{ J s}$ $h = 4.14 \times 10^{-15} \text{ eV s}$
16	speed of light	$c = 3.0 \times 10^8 \text{ m s}^{-1}$

Detailed study 3.1 – Synchrotron and applications

17	energy transformations for electrons in an electron gun (<100 keV)	$\frac{1}{2} m v^2 = eV$
18	radius of electron beam	$r = p/eB$
19	force applied to an electron beam	$F = evB$
20	Bragg's law	$n\lambda = 2d \sin \theta$
21	electric field between charged plates	$E = \frac{V}{d}$

Detailed study 3.2 – Photonics

22	band gap energy	$E = \frac{hc}{\lambda}$
23	Snell's law	$n_1 \sin i = n_2 \sin r$

Detailed study 3.3 – Sound

24	speed, frequency and wavelength	$v = f\lambda$
25	intensity and levels	sound intensity level $(\text{in dB}) = 10 \log_{10} \left(\frac{I}{I_0} \right)$ where $I_0 = 1.0 \times 10^{-12} \text{ W m}^{-2}$

Prefixes/Unitsp = pico = 10^{-12} n = nano = 10^{-9} μ = micro = 10^{-6} m = milli = 10^{-3} k = kilo = 10^3 M = mega = 10^6 G = giga = 10^9 t = tonne = 10^3 kg **END OF DATA SHEET**